1 Introduction

The purpose of this lab is to introduce second-order circuits and investigate the effects of inductances and capacitances in a digital circuit. The lab focuses on building an inductor and analyzing its characteristics, as well as reducing ringing in a digital circuitry. The equipment used includes a breadboard, power supply, multimeter, signal generator, oscilloscope, and various components such as resistors, capacitors, insulated solenoid wire, and a ferrite core.

2 Preliminary Work

2.1 Task 1

In the preliminary work, we find the inductance of a the coil using the formula given below that is needed to calculate the inductance

- $$L = \frac{\mu N^2 A}{l}$$

  (Where μ is $\mu_0 * \mu_r$)

  1. L is inductance of coil in Henrys
  2. N is number of turns in wire coil
  3. μ is permeability of core material
  4. μ_r is Relative permeability (Dimensionless) ($\mu_0 = 1$ for air)
  5. μ_0 is
  6. A is Area of coil in square meters $= \pi * r^2$
  7. l is average length of coil in meters

- Now we build an inductor such that the inductance is 10 micro-henry with given values of ferrit core in the figure 1.

Table 1: Ferrite core specs

Core diameter	8 mm
Core Length	45 mm
Relative Permeability	50 (±%15)
Solenoid wire diameter	0.5 mm

Figure 1: Ferrite core specs.

- so here we have
  $\mu_0 = 4 * \pi * 10^{-7}$
  $\mu_r = 1$

- so we have

$$L = \frac{10 * 10^{-6} 4 * \pi * 10^{-7} * 50 * N^2 4.5 * 10^{-3} * \pi}{45 * 10}$$

- According to the information provided, the value "50" represents the relative permeability of the ferrite core, which indicates that the permeability (μ) of the ferrite material is 50 times larger than the permeability of free space (μ_0). As a result, the equation for calculating the inductance needs to be multiplied by 50. Additionally, the wire size is taken into account when determining the cross-sectional area of the coil. Using a numerical solver, it is determined that the required number of turns (N) to achieve an inductance of 10 $\mu * H$ is approximately 10.6, which can be rounded up to 11 turns. Therefore, the coil should be wrapped with 11 turns of wire around the ferrite core to achieve the desired inductance.

2.2 Task 2

With connecting inductor in series with a known resistor t oa sinusoidal voltage we can find inductance. The impedance of the inductor can be calculated by measuring the voltage drop across it and contrasting it with the voltage across the resistor. The inductance of the inductor can then be determined using this impedance value.

- Equation of impedance is $X_L = 2j\pi fL$

3 Lab Tasks

3.1 1. Constructing the Inductor

To design an inductor with an inductance of 10 microhenries (μH), we will build our own inductor coil by wrapping an insulated wire around a ferrite core. The specifications of the ferrite core are as follows:

1. Core diameter: 8 mm
2. Core length: 45 mm
3. Relative permeability: 50 ($\pm 15\%$)
4. Solenoid wire diameter: 0.5 mm
5. By conducting a literature search, we can calculate the inductance of a coil using the following formula:

$$L = \frac{\mu N^2 A}{l}$$

(Where μ is $\mu_0 * \mu_r$)

6. We have calculated inductor in the preliminary work with ferrit core with 0.5 mm thick enwrapped 11 turns of copper wire is resulted creating 9.48μH inductance.

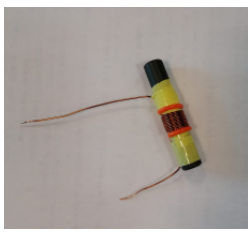


Figure 2: Inductor.

7. After enwrapping the wire on the ferrit core, we have cleaned legs of wire on both sides to get rid of paint.

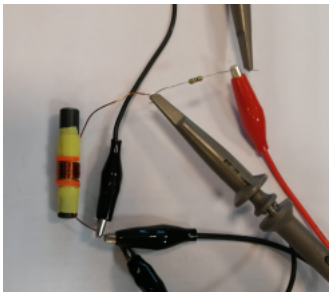


Figure 3: Resistor and Inductor.

8. Then we connected Inductor to 10Ω resistor in series as shown in Figure 3.

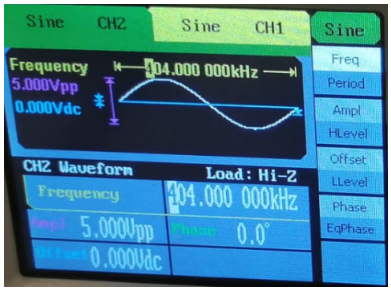


Figure 4: Function wave generator.

9. We have connected the RL circuit to the Wave generator as shown in Figure 4 and we gave a sinusoidal wave signal to the circuit with 404KHz with $5V_{pp}$ as shown in figure 4.

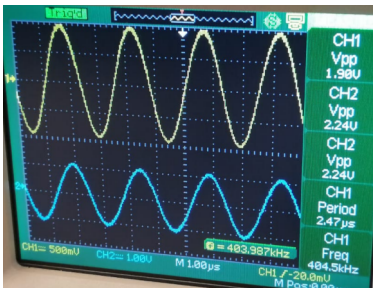


Figure 5: Oscilloscope values for RL circuit.

10. When we analyze the wave output from oscilloscope we obtain outputs $1.9V_{pp}$for output and $2.24V_{pp}$ for input as shown Figure 5.

3.2 Measuring the Inductance

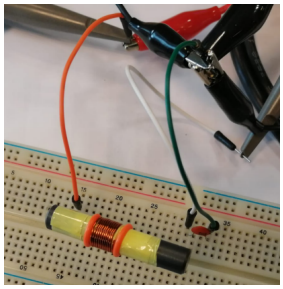


Figure 6: Inductor and Conductor circuit.

1. Firstly we have built the LC circuit given in the lab instructions and connected the circuit to the wave generator as shown in figure 6.

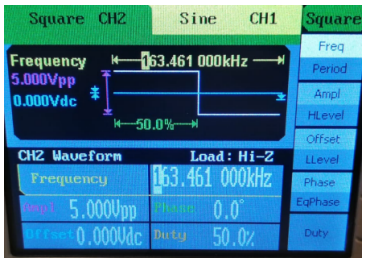


Figure 7: Wave generator connected to the LC circuit.

2. we gave an square wave with $5V_{pp}$ and 163.461 KHz because of calculated resonance as shown in Figure 7.
   formula of resonance is

$$f = \frac{1}{2\pi\sqrt{CL}} = \frac{1}{2\pi\sqrt{100 * 10^{-9} * 9.48 * 10^{-6}}}$$

   gives 163.461KHz



Figure 8: Ferrite core specs.

3. When we connect the circuit to the oscilloscope we obtain the figure 8. On the oscilloscope we can obtain CH1 is input with 5.76 V_{pp} and output is CH2 with 1.48 V_{pp}

4 Conclusion

In this lab, we successfully constructed an inductor with a goal inductance of 10 muH by encircling a ferrite core 11 times with 0.5 mm insulated wire. We measured the voltage drop across the inductor after connecting it in series with a 10 resistor.

We evaluated the waveform after applying a sinusoidal wave signal with a frequency of 404 KHz and 5 Vpp and measured the voltage at 1.9 Vpp for the output and 2.24 Vpp for the input. These outcomes enabled us to determine the impedance of the inductor and confirm the inductance value.

Additionally, we built an LC circuit and chose a square wave with a resonance frequency of 163.461 KHz and a voltage of 5 Vpp. An output voltage of 1.48 Vpp and an input voltage of 5.76 Vpp were both visible on the oscilloscope.

The implications of inductances and capacitances in digital circuits, second-order circuits, and the practical elements of creating and analyzing inductors were all covered in depth in this lab.